Radical Axes

Problem 1. A point C lies on a line segment AB between A and B and circles are drawn having AC and CB as diameters. A common tangent line to both circles touches the circle with AC as diameter at $P \neq C$ and the circle with CB as diameter at $Q \neq C$. Prove that lines AP, BQ and the common tangent line to both circles at C all meet at a single point which lies on the circle with AB as a diameter.

Problem 2. In acute triangle ABC, let D, E, F denote the feet of the altitudes from A, B, C, respectively, and let ω be the circumcircle of $\triangle AEF$. Let ω_1 and ω_2 be the circles through D tangent to ω at E and F, respectively. Show that ω_1 and ω_2 meet at a point P on BC other than D.

Problem 3. In a convex quadrilateral $ABCD$, $\angle ABC = \angle ADC < 90°$. Let E be a point on the side AB such that $CB = CE$. Let F be a point on the side AD such that $CD = CF$. Prove that C, the circumcentre of $\triangle ABD$ and the circumcentre of $\triangle CEF$ are collinear.

Problem 4. Suppose AB is a chord of $\odot O$. Let M be the midpoint of arc AB. Given an arbitrary point C outside $\odot O$, construct tangents CS and CT to $\odot O$ where S and T are the contact points. The lines MS and MT meet AB at E and F respectively. The lines through E and F perpendicular to AB intersect OS and OT at X and Y respectively. A line through C intersects $\odot O$ at two distinct points P and Q. The line MP meets AB at R. Let Z be the circumcentre of $\triangle PQR$. Prove that X, Y, Z are collinear.

Problem 5. Diagonals of a cyclic quadrilateral $ABCD$ intersect at point K. The midpoints of diagonals AC and BD are M and N, respectively. The circumscribed circles ADM and BCM intersect at points M and L. Prove that the points K, L, M, and N lie on a circle. (All points are supposed to be different.)